

Fengping River Hydropower — Main Analysis

OVERVIEW

This program evaluates the financial feasibility of the proposed Fengping River hydropower project under historic flows and two climate change scenarios (SSP2-4.5 and SSP5-8.5).

Program contract

- **Inputs:** daily streamflow record (1958–2025), sediment rating-curve data, weir design specifications, climate projection parameters, finance parameters (FIT, CAPEX, LTV, loan rate, equity rate, project lifetime).
- **Outputs:** annual energy generation (GWh), annual revenue (NTD), 20-year NPV, and an NPV feasibility frontier across the loan-rate $\times$ LTV parameter grid, reported for each scenario (baseline, SSP2-4.5, SSP5-8.5).
- **Parameters:** e-flow mode, sediment toggle, number of bootstrap simulations, projection year, finance parameter ranges.

MODULES

Module	Purpose	Reuse
hydro_reservoir.R	Reservoir simulation: flow to energy	Used 3 times (baseline, SSP2-4.5, SSP5-8.5)
hydro_sediment.R	Sediment load to trap efficiency to storage loss	1 time
hydro_climate.R	Bootstrap climate scenarios from historic flow	Used 2 times (SSP2-4.5, SSP5-8.5)
hydro_finance.R	Energy to revenue to NPV to feasibility frontier	Used 3 times (one per scenario)
hydro_plot.R	Standardized plots for each output	many
hydro_test.R	Unit tests for all modules	1 time at setup

SETUP

PACKAGES

```
packages <- c(
  "tidyverse", "here", "ggplot2",
  "patchwork", "scales", "lubridate", "purrr"
)
installed_packages <- rownames(installed.packages())
for (pkg in packages) {
  if (!pkg %in% installed_packages) install.packages(pkg)
}
invisible(lapply(packages, library, character.only = TRUE))
```

Loading Modules

```
source(here("module", "hydro_reservoir.R"))
source(here("module", "hydro_sediment.R"))
source(here("module", "hydro_finance.R"))
source(here("module", "hydro_climate.R"))
source(here("module", "hydro_plot.R"))
source(here("module", "hydro_test.R"))

# Run all tests first - stops execution if anything fails
run_all_tests()
```

Loading Data

```
daily_flow <- read_csv(here("data", "lishan_daily_clean.csv"),
  show_col_types = FALSE)
decadal_flow <- read_csv(here("data", "lishan_decadal_long.csv"),
  show_col_types = FALSE)
sed_clean <- read_csv(here("data", "sediment_clean.csv"),
  show_col_types = FALSE)
weir_design <- read_csv(here("data", "weir_design_flow.csv"),
  show_col_types = FALSE)
rc_params <- read_csv(here("data", "rating_curve_params.csv"),
  show_col_types = FALSE)
```

Financial Parameters - adjust to explore different scenarios

```
# Feed-In Tariff (NTD / kWh)
# Source:      114      (> 20,000 kW class)
FIT_NTD_PER_KWH <- 2.8599
# Total CAPEX (NTD)
# Source: Yongwei investor conference 2025 (up from NTD 6.4B due to inflation)
CAPEX_NTD <- 9.7e9
# Loan-to-value ratio (fraction)
# Typical range for Taiwan utility-scale renewables: 0.70 - 0.80
LTV <- 0.75
# Annual loan interest rate (fraction)
# Current Taiwan syndicated loan range: 0.01 - 0.05
R_LOAN <- 0.03
# Equity discount rate (fraction)
# Required return for private utility investor
R_EQUITY <- 0.08
# Project lifetime (years)
# Source: Shihfeng Power Co. EIA documents
T_YEARS <- 35L

# Sensitivity grid ranges
# Adjust these to zoom in / out on the feasibility frontier
R_LOAN_RANGE <- seq(0.01, 0.05, by = 0.005) # 1% to 5%
LTV_RANGE    <- seq(0.50, 0.90, by = 0.05)  # 50% to 90%

cat("=== Finance parameters ===\n")
```

=== Finance parameters ===

```
cat(sprintf("FIT rate      : NTD %.4f / kWh\n", FIT_NTD_PER_KWH))
```

FIT rate : NTD 2.8599 / kWh

```
cat(sprintf("CAPEX        : NTD %.1f billion\n", CAPEX_NTD / 1e9))
```

CAPEX : NTD 9.7 billion

```
cat(sprintf("LTV           : %.0f%%\n", LTV * 100))
```

LTV : 75%

```
cat(sprintf("Loan rate      : %.1f%%\n", R_LOAN * 100))
```

Loan rate : 3.0%

```
cat(sprintf("Equity rate   : %.1f%%\n", R_EQUITY * 100))
```

Equity rate : 8.0%

```
cat(sprintf("Project life : %d years\n", T_YEARS))
```

Project life : 35 years

Baseline (historic) scenarios

Reservoir simulation

```
# Run reservoir simulation - baseline, recommended e-flow
sim_baseline <- run_reservoir_simulation(
  daily_flow = daily_flow,
  e_flow_mode = "recommended"
)
```

simulate_one_weir [W1]: net_head estimated as 86.3 m

simulate_one_weir [W2]: net_head estimated as 349.4 m

```
cat("=== Baseline simulation (1958-2025, recommended e-flow) ===\n")
```

=== Baseline simulation (1958-2025, recommended e-flow) ===

```
cat("Annual energy W1 (GWh) :",  
    round(sum(sim_baseline$W1$energy_kwh) / 1e6, 1), "\n")
```

Annual energy W1 (GWh) : 1683.8

```
cat("Annual energy W2 (GWh) :",  
    round(sum(sim_baseline$W2$energy_kwh) / 1e6, 1), "\n")
```

Annual energy W2 (GWh) : 5437.6

```
cat("E-flow satisfied W1 (%) :",  
    round(mean(sim_baseline$W1$eflow_satisfied) * 100, 1), "%\n")
```

E-flow satisfied W1 (%) : 40.2 %

```
cat("E-flow satisfied W2 (%) :",  
    round(mean(sim_baseline$W2$eflow_satisfied) * 100, 1), "%\n")
```

E-flow satisfied W2 (%) : 91.7 %

Reservoir Operation Plots

```
# Default: first 90 days  
plot_reservoir_operation(  
  sim_weir = sim_baseline$W1,  
  weir_label = "W1",  
  e_flow_mode = "recommended"  
)
```

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (1958-10-01 to 1958-12-
29)' in
'mbcsToSbcs': dot substituted for <e2>

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conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <95>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

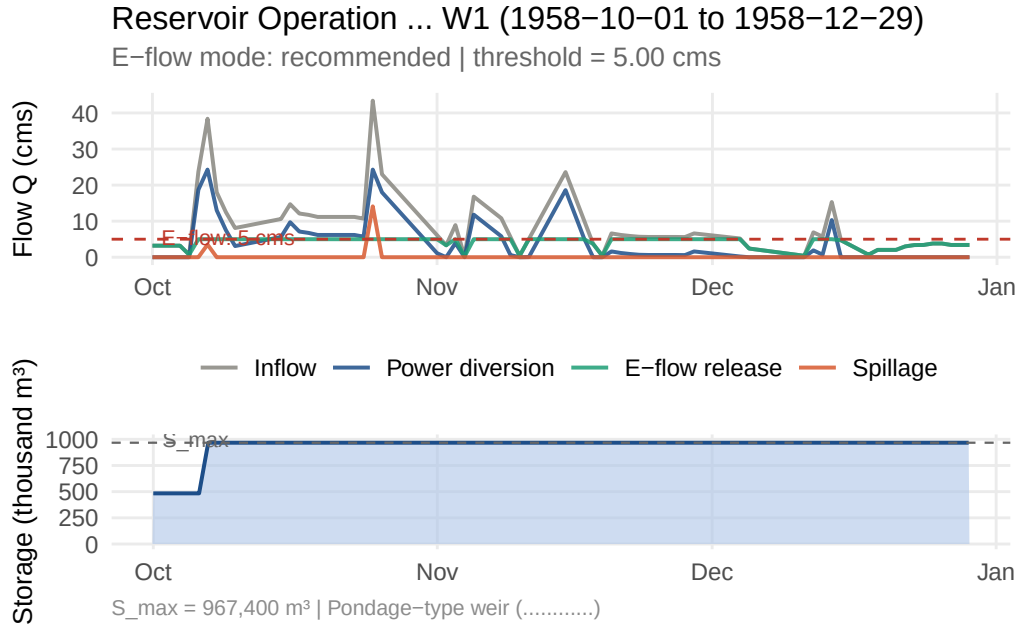
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e5>

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ( )' in
'mbcsToSbcs': dot substituted for <bc>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ( )' in
'mbcsToSbcs': dot substituted for <8f>
```



```
# Typhoon season: July-September 2023
plot_reservoir_operation(
  sim_weir      = sim_baseline$W1,
  date_start    = "2023-07-01",
  date_end      = "2023-09-30",
  weir_label    = "W1",
  e_flow_mode   = "recommended"
)
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-
30)' in
'mbcsToSbcs': dot substituted for <e2>
```

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-
30)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-
30)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-
30)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-
30)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-
30)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-
30)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-
30)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-
30)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e8>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <aa>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bf>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <95>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e5>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bc>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <8f>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e8>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <aa>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bf>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <95>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e5>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bc>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <8f>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e8>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <aa>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bf>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <95>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e5>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bc>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <8f>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-
30)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-
30)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-
30)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-30)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-30)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-30)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-30)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-30)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-07-01 to 2023-09-30)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e8>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <aa>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bf>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <95>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e5>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bc>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <8f>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e8>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <aa>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bf>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <95>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e5>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bc>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <8f>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e8>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <aa>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bf>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <95>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

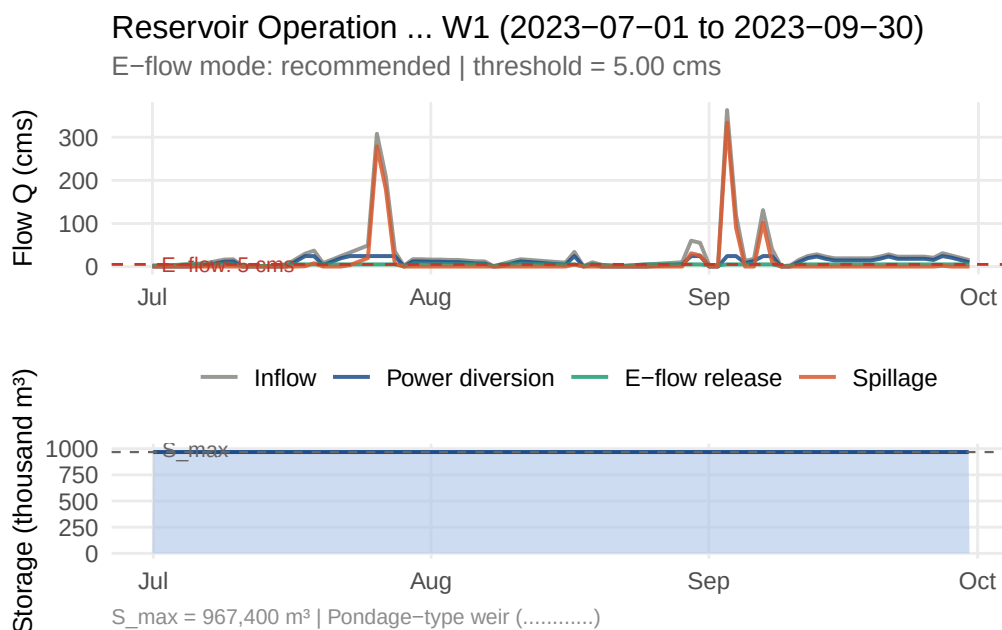
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>


```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ( )' in
'mbcsToSbcs': dot substituted for <e5>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ( )' in
'mbcsToSbcs': dot substituted for <bc>
```

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ( )' in
'mbcsToSbcs': dot substituted for <8f>
```



```
# Dry season: January-March 2023
plot_reservoir_operation(
  sim_weir    = sim_baseline$W1,
  date_start  = "2023-01-01",
  date_end    = "2023-03-31",
  weir_label  = "W1",
  e_flow_mode = "recommended"
)
```

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-31)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e8>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <aa>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bf>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <95>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e5>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bc>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <8f>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e8>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <aa>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
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Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
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Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
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Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e5>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bc>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <8f>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e8>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
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Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bf>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <95>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>

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conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
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Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bc>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <8f>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-31)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-31)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
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'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
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'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Reservoir Operation - W1 (2023-01-01 to 2023-03-
31)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e8>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <aa>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bf>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <95>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e5>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bc>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <8f>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e8>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <aa>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bf>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <95>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e5>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bc>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <8f>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e8>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <aa>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bf>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <95>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b4>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e6>

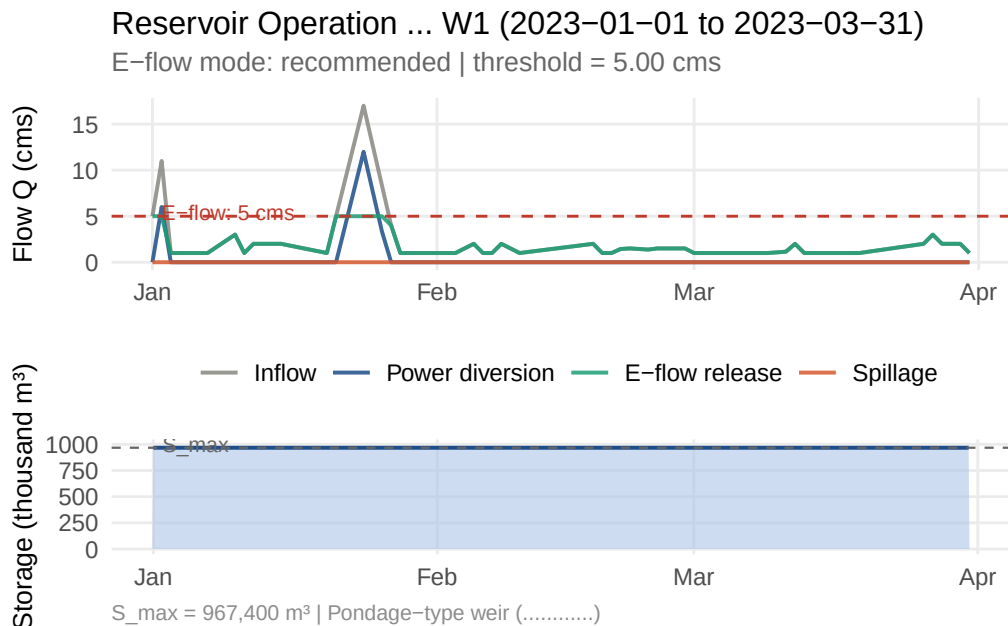
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <b1>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <a0>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <e5>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <bc>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'S_max = 967,400 m³ | Pondage-type weir ()' in
'mbcsToSbcs': dot substituted for <8f>



Sediment

```
sed_out <- build_sediment_outputs(  
  sed_csv_path = here("data", "sediment_clean.csv"),  
  daily_flow   = daily_flow,  
  plot_fit     = FALSE  
)
```

Rating curve fitted: $SSL = 11.5178 \times Q^{1.5085}$ | $R^2 = 0.638$ | $n = 492$

Annual SSL estimate (full daily record): 19963784 MT/year

```
cat("\n=== Sediment summary ===\n")
```

=== Sediment summary ===

```
cat("Annual SSL (MT/year):", round(sed_out$annual_ssl_mt), "\n")
```

Annual SSL (MT/year): 19963784

```
cat("\nTrap efficiency W1 (year 1, 10, 35):\n")
```

Trap efficiency W1 (year 1, 10, 35):

```
sed_out$trap_W1 |>  
  filter(year %in% c(1, 10, 35)) |>  
  select(year, S_remaining_m3, TE) |>  
  print()
```

	year	S_remaining_m3	TE
1	1	0	0.3754
2	10	0	0.0000
3	35	0	0.0000

Energy Generation Calendar

```
#  
plot_energy_calendar_heatmap(sim_baseline$W1, year_filter = 2010)
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':  
dot substituted for <e2>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':  
dot substituted for <80>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':  
dot substituted for <94>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':  
dot substituted for <e2>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':  
dot substituted for <80>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':  
dot substituted for <94>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':  
dot substituted for <e2>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':  
dot substituted for <80>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':  
dot substituted for <94>
```

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
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Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
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conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
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Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
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Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
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conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
dot substituted for <e2>

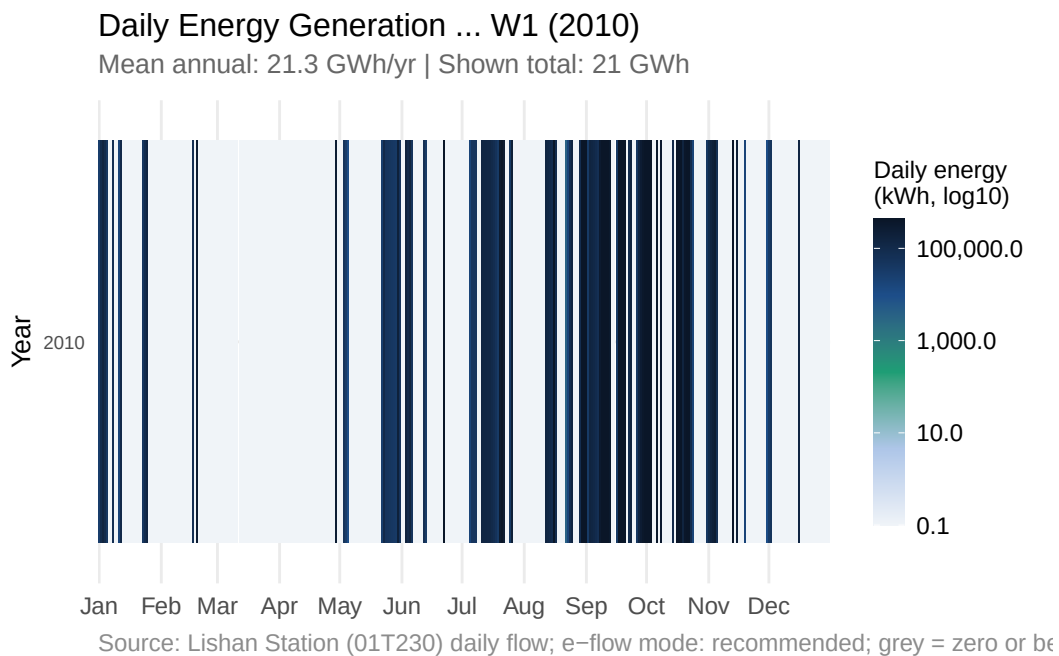
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
dot substituted for <94>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
dot substituted for <80>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010)' in 'mbcsToSbcs':
dot substituted for <94>



```
#  
plot_energy_calendar_heatmap(sim_baseline$W1, year_filter = 2010:2020)
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in  
'mbcsToSbcs': dot substituted for <e2>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in  
'mbcsToSbcs': dot substituted for <80>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in  
'mbcsToSbcs': dot substituted for <94>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in  
'mbcsToSbcs': dot substituted for <e2>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in  
'mbcsToSbcs': dot substituted for <80>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in  
'mbcsToSbcs': dot substituted for <93>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in  
'mbcsToSbcs': dot substituted for <e2>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in  
'mbcsToSbcs': dot substituted for <80>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :  
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in  
'mbcsToSbcs': dot substituted for <94>
```


Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <94>

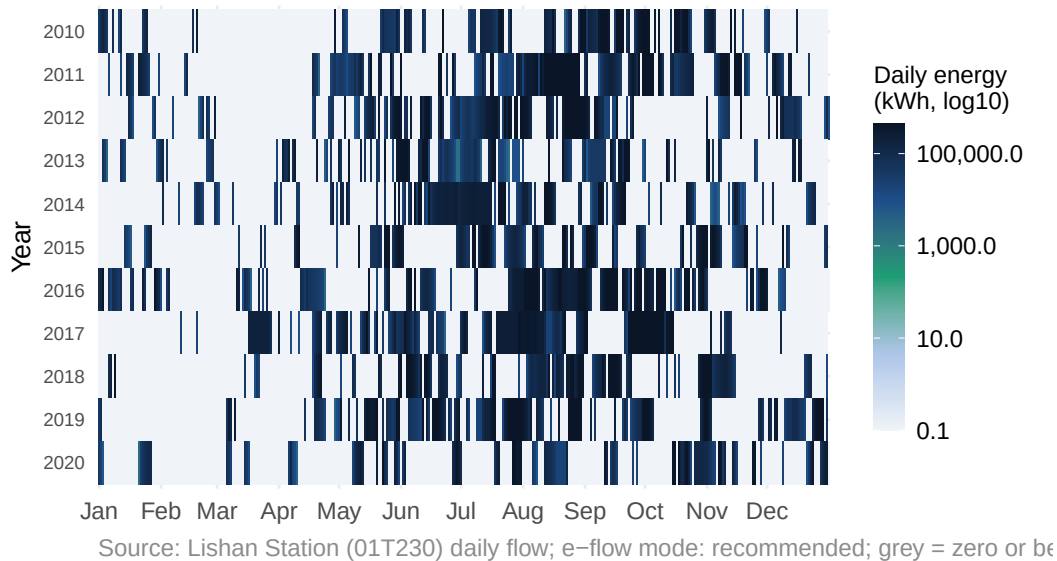
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (2010-2020)' in
'mbcsToSbcs': dot substituted for <93>

Daily Energy Generation ... W1 (2010...2020)

Mean annual: 23.5 GWh/yr | Shown total: 259 GWh



```
#
plot_energy_calendar_heatmap(sim_baseline$W1)
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <e2>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <80>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <94>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <e2>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <80>
```

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <94>

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conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <94>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <94>

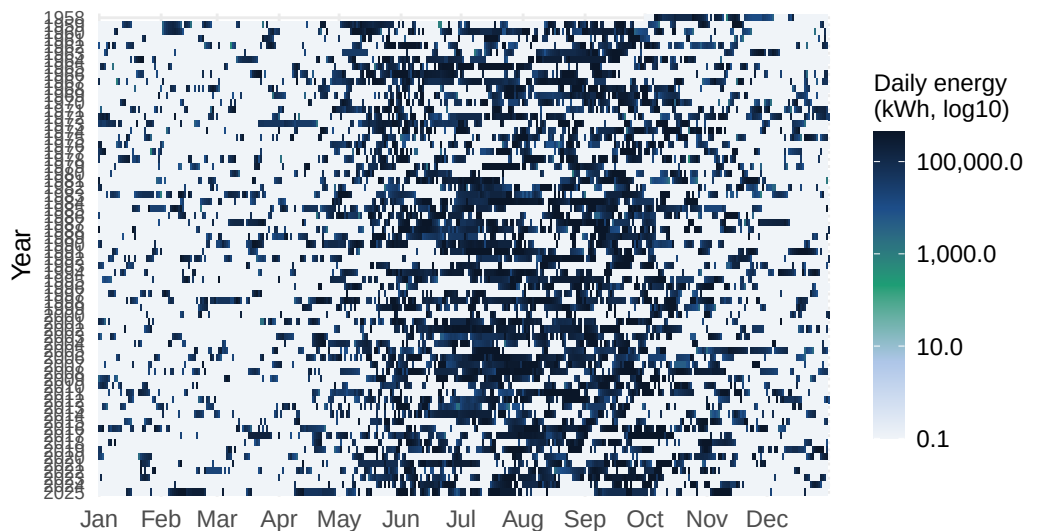
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Daily Energy Generation - W1 (All Years)' in
'mbcsToSbcs': dot substituted for <94>

Daily Energy Generation ... W1 (All Years)

Mean annual: 24.8 GWh/yr | Shown total: 1684 GWh



Source: Lishan Station (01T230) daily flow; e-flow mode: recommended; grey = zero or below

Finance — baseline without sediment

```
total_gwh_baseline <- (sum(sim_baseline$W1$energy_kwh) +
                        sum(sim_baseline$W2$energy_kwh)) / 1e6

fin_no_sed <- build_finance_outputs(
  annual_gwh      = total_gwh_baseline,
  scenario_label = "Baseline (no sediment)"
)
```

[Baseline (no sediment)] Revenue: NTD 20.37B/yr | NPV: NTD 229.86B | IRR: NA% | Payback: 1 yr

Finance — baseline with sediment adjustment

```
# Sediment reduces storage → reduces power output over time
# Use mean TE over project life to adjust annual GWh
mean_te_W1 <- mean(sed_out$trap_W1$TE)
mean_te_W2 <- mean(sed_out$trap_W2$TE)
```



```
# Approximate: energy loss proportional to storage loss
# (simplified - full dynamic coupling in research version)
storage_loss_factor <- 1 - (mean_te_W1 * 0.6 + mean_te_W2 * 0.4)
gwh_with_sed <- total_gwh_baseline * storage_loss_factor

fin_with_sed <- build_finance_outputs(
  annual_gwh      = gwh_with_sed,
  scenario_label = "Baseline (with sediment)"
)
```

[Baseline (with sediment)] Revenue: NTD 20.15B/yr | NPV: NTD 227.35B | IRR: NA% | Payback: 1

```
cat("=== Finance comparison ===\n")
```

```
=== Finance comparison ===
```

```
cat(sprintf("Without sediment: NTD %.2fB/yr | NPV NTD %.2fB\n",
  fin_no_sed$annual_revenue_ntd / 1e9,
  fin_no_sed$npv_ntd / 1e9))
```

Without sediment: NTD 20.37B/yr | NPV NTD 229.86B

```
cat(sprintf("With sediment : NTD %.2fB/yr | NPV NTD %.2fB\n",
  fin_with_sed$annual_revenue_ntd / 1e9,
  fin_with_sed$npv_ntd / 1e9))
```

With sediment : NTD 20.15B/yr | NPV NTD 227.35B

Baseline Feasibility Frontier

```
# Sensitivity for baseline scenario
sens_no_sed <- fin_no_sed$sensitivity
sens_with_sed <- fin_with_sed$sensitivity

sens_combined <- bind_rows(sens_no_sed, sens_with_sed)
user_point <- data.frame(
  scenario = c("Baseline (no sediment)", "Baseline (with sediment)"),
```

```

    r_loan_pct = R_LOAN * 100,
    ltv_pct     = LTV * 100
)

plot_npv_feasibility_frontier(sens_combined) +
  geom_point(
    data      = user_point,
    aes(x = r_loan_pct, y = ltv_pct),
    shape     = 23,
    size      = 4,
    fill      = "white",
    colour    = "black",
    stroke    = 1.2
  ) +
  labs(caption = paste0(
    "Diamond = current parameter setting",
    " (LTV = ", LTV * 100, "%, rate = ", R_LOAN * 100, "%)\n",
    "CAPEX = NTD ", CAPEX_NTD / 1e9, "B; ",
    "FIT = NTD ", FIT_NTD_PER_KWH, "/kWh; ",
    "Equity rate = ", R_EQUITY * 100, "%"
  ))

```

Warning: `stat\_contour()`: Zero contours were generated

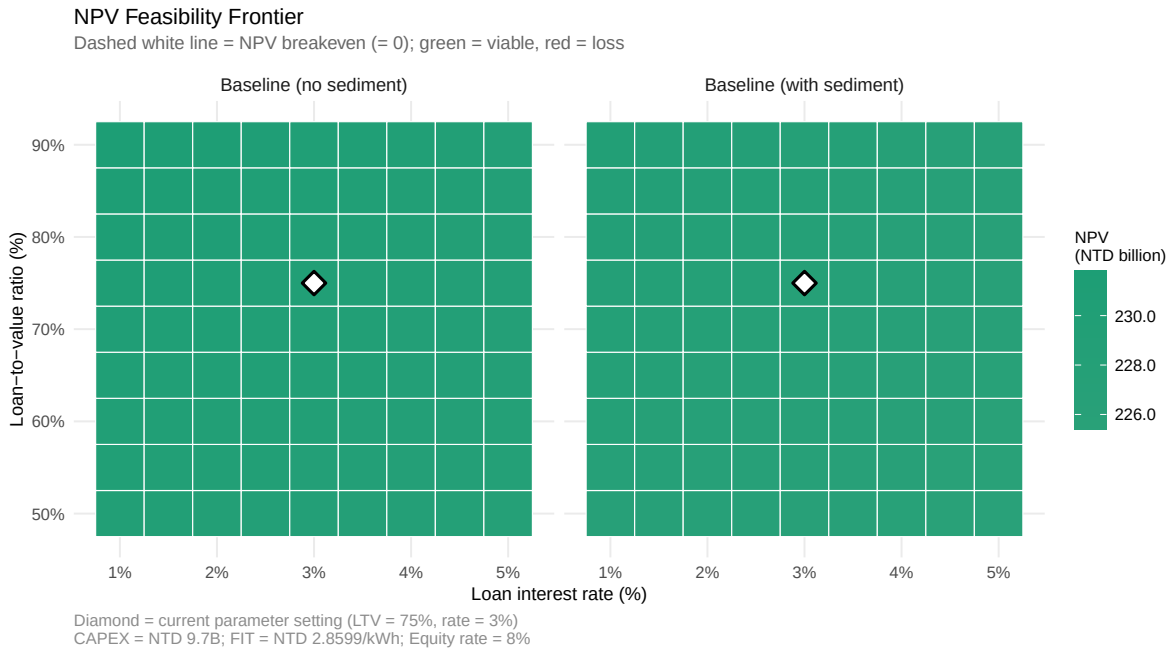
Warning in min(x): no non-missing arguments to min; returning Inf

Warning in max(x): no non-missing arguments to max; returning -Inf

Warning: `stat\_contour()`: Zero contours were generated

Warning in min(x): no non-missing arguments to min; returning Inf

Warning in max(x): no non-missing arguments to max; returning -Inf



CLIMATE CHANGE SCENARIOS

Generate scenarios

```
climate_out <- build_climate_scenarios(
  daily_flow = daily_flow, # data.frame with columns: date, Q_cms
  proj_year  = 2065L,      # adjust to your target year
  n_sim      = 200L,       # 200 for assignment; 1000 for publication
  seed       = 42L
)
scenario_names <- names(climate_out)
scenario_finance <- vector("list", length(scenario_names))

for (i in seq_along(scenario_names)) {

  ssp_name <- scenario_names[[i]]
  scen     <- climate_out[[ssp_name]]

  # P50 mean flow → rough annual GWh
  p50_gwh <- scen$summary$percentiles |>
    filter(variable == "mean_Q") |>
```

```

    pull(P50) * 365 * 86400 / 1e9

    fin <- build_finance_outputs(
      annual_gwh      = p50_gwh,
      scenario_label = ssp_name
    )

    scenario_finance[[i]] <- data.frame(
      scenario      = ssp_name,
      gwh_p50       = p50_gwh,
      npv_b_ntd     = fin$npv_ntd / 1e9,
      irr_pct       = fin$irr_pct,
      payback_yr    = fin$payback_yr
    )
  }

scenario_finance <- bind_rows(scenario_finance)
print(scenario_finance)

```

	scenario	gwh_p50	npv_b_ntd	irr_pct	payback_yr
1	baseline	0.3066878	-7.491197	NA	Inf
2	ssp245	0.3349030	-7.490257	NA	Inf
3	ssp585	0.3772259	-7.488846	NA	Inf

NPV across scenarios (loop over SSPs)

This is the program's main for loop: for each climate scenario, run finance on the P50 mean-flow projection and collect outputs.

```

# Run finance for each scenario P50
scenario_finance <- purrr::imap_dfr(climate_out, function(scen, ssp_name) {

  p50_gwh <- scen$summary$percentiles |>
    filter(variable == "mean_Q") |>
    pull(P50) * 365 * 86400 / 1e9 # rough GWh from mean Q

  fin <- build_finance_outputs(
    annual_gwh      = p50_gwh,
    scenario_label = ssp_name
  )

```

```

data.frame(
  scenario    = ssp_name,
  gwh_p50     = p50_gwh,
  npv_b_ntd   = fin$npv_ntd / 1e9,
  irr_pct     = fin$irr_pct,
  payback_yr  = fin$payback_yr
)
})

```

[baseline] Revenue: NTD 0.00B/yr | NPV: NTD -7.49B | IRR: NA% | Payback: Inf yr

[ssp245] Revenue: NTD 0.00B/yr | NPV: NTD -7.49B | IRR: NA% | Payback: Inf yr

[ssp585] Revenue: NTD 0.00B/yr | NPV: NTD -7.49B | IRR: NA% | Payback: Inf yr

```
print(scenario_finance)
```

	scenario	gwh_p50	npv_b_ntd	irr_pct	payback_yr
1	baseline	0.3066878	-7.491197	NA	Inf
2	ssp245	0.3349030	-7.490257	NA	Inf
3	ssp585	0.3772259	-7.488846	NA	Inf

Feasibility Frontier - Climate Scenarios

```

sens_no_sed_clim <- run_sensitivity(
  annual_revenue_ntd = fin_no_sed$annual_revenue_ntd,
  r_loan_seq         = R_LOAN_RANGE,
  ltv_seq             = LTV_RANGE,
  scenario_label      = "Without sediment"
)

sens_with_sed_clim <- run_sensitivity(
  annual_revenue_ntd = fin_with_sed$annual_revenue_ntd,
  r_loan_seq         = R_LOAN_RANGE,
  ltv_seq             = LTV_RANGE,
  scenario_label      = "With sediment"
)

```

```

sens_combined_clim <- bind_rows(sens_no_sed_clim, sens_with_sed_clim)

user_point_clim <- data.frame(
  scenario_label = c("Without sediment", "With sediment"),
  r_loan_pct     = R_LOAN * 100,
  ltv_pct        = LTV * 100
)

plot_npv_feasibility_frontier(sens_combined_clim) +
  geom_point(
    data      = user_point_clim,
    aes(x = r_loan_pct, y = ltv_pct),
    shape     = 23,
    size      = 4,
    fill      = "white",
    colour    = "black",
    stroke    = 1.2,
    inherit.aes = FALSE
  ) +
  labs(caption = paste0(
    "Diamond = current parameter setting",
    " (LTV = ", LTV * 100, "%, rate = ", R_LOAN * 100, "%)\n",
    "CAPEX = NTD ", CAPEX_NTD / 1e9, "B; ",
    "FIT = NTD ", FIT_NTD_PER_KWH, "/kWh; ",
    "Equity rate = ", R_EQUITY * 100, "%"
  ))

```

Warning: `stat\_contour()`: Zero contours were generated

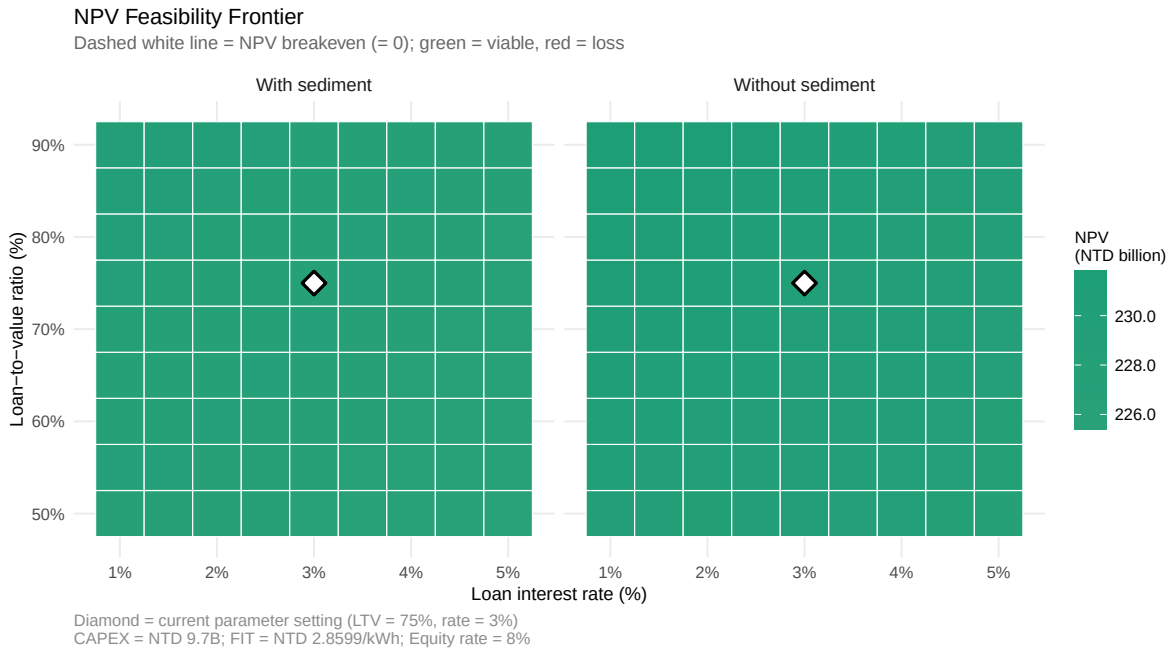
Warning in min(x): no non-missing arguments to min; returning Inf

Warning in max(x): no non-missing arguments to max; returning -Inf

Warning: `stat\_contour()`: Zero contours were generated

Warning in min(x): no non-missing arguments to min; returning Inf

Warning in max(x): no non-missing arguments to max; returning -Inf



OPERATIONS COMPARISON

Forecast-informed simulation (24-hour lookahead)

```
sim_forward24 <- run_reservoir_simulation_forward24(
  daily_flow = daily_flow,
  e_flow_mode = "recommended",
  scale_w2 = TRUE
)

annual_forward24 <- summarise_annual_generation(sim_forward24)

result <- annual_forward24 |>
  summarise(
    mean_energy_gwh = mean(energy_gwh_total, na.rm = TRUE),
    p10_energy_gwh = quantile(energy_gwh_total, 0.10, na.rm = TRUE),
    p90_energy_gwh = quantile(energy_gwh_total, 0.90, na.rm = TRUE),
    mean_revenue_b_ntd = mean(revenue_ntd, na.rm = TRUE) / 1e9,
    mean_spill_m3 = mean(spill_m3_W1 + spill_m3_W2, na.rm = TRUE)
  )
print(result)
```

```
# A tibble: 1 x 5
  mean_energy_gwh p10_energy_gwh p90_energy_gwh mean_revenue_b_ntd mean_spill_m3
      <dbl>         <dbl>         <dbl>         <dbl>         <dbl>
1         48.2         31.0         62.0         0.138        95554897.
```

Finance — baseline without sediment

```
# Finance - without sediment
fin_no_sed <- build_finance_outputs(
  annual_gwh      = total_gwh_baseline,
  scenario_label  = "Without sediment",
  fit_ntd_per_kwh = FIT_NTD_PER_KWH,
  ltv             = LTV,
  r_loan         = R_LOAN
)
```

[Without sediment] Revenue: NTD 20.37B/yr | NPV: NTD 229.86B | IRR: NA% | Payback: 1 yr

Finance — baseline with sediment adjustment

```
## Finance - with sediment
fin_with_sed <- build_finance_outputs(
  annual_gwh      = gwh_with_sed,
  scenario_label  = "With sediment",
  fit_ntd_per_kwh = FIT_NTD_PER_KWH,
  ltv             = LTV,
  r_loan         = R_LOAN
)
```

[With sediment] Revenue: NTD 20.15B/yr | NPV: NTD 227.35B | IRR: NA% | Payback: 1 yr

```
# Sensitivity across rate × LTV grid
sens_no_sed <- run_sensitivity(
  annual_revenue_ntd = fin_no_sed$annual_revenue_ntd,
  r_loan_seq         = R_LOAN_RANGE,
  ltv_seq            = LTV_RANGE,
  scenario_label     = "Without sediment"
)
```



```

sens_with_sed <- run_sensitivity(
  annual_revenue_ntd = fin_with_sed$annual_revenue_ntd,
  r_loan_seq         = R_LOAN_RANGE,
  ltv_seq            = LTV_RANGE,
  scenario_label      = "With sediment"
)

sens_combined <- bind_rows(sens_no_sed, sens_with_sed)

# Mark user's chosen parameters on the plot
user_point <- data.frame(
  scenario    = c("Without sediment", "With sediment"),
  r_loan_pct  = R_LOAN * 100,
  ltv_pct     = LTV * 100
)

plot_npv_feasibility_frontier(sens_combined) +
  geom_point(
    data      = user_point,
    aes(x = r_loan_pct, y = ltv_pct),
    shape     = 23,
    size      = 4,
    fill      = "white",
    colour    = "black",
    stroke    = 1.2
  ) +
  labs(caption = paste0(
    "Diamond = current parameter setting",
    " (LTV = ", LTV * 100, "%, rate = ", R_LOAN * 100, "%)\n",
    "CAPEX = NTD ", CAPEX_NTD / 1e9, "B; ",
    "FIT = NTD ", FIT_NTD_PER_KWH, "/kWh; ",
    "Equity rate = ", R_EQUITY * 100, "%"
  ))

```

Warning: `stat\_contour()`: Zero contours were generated

Warning in min(x): no non-missing arguments to min; returning Inf

Warning in max(x): no non-missing arguments to max; returning -Inf

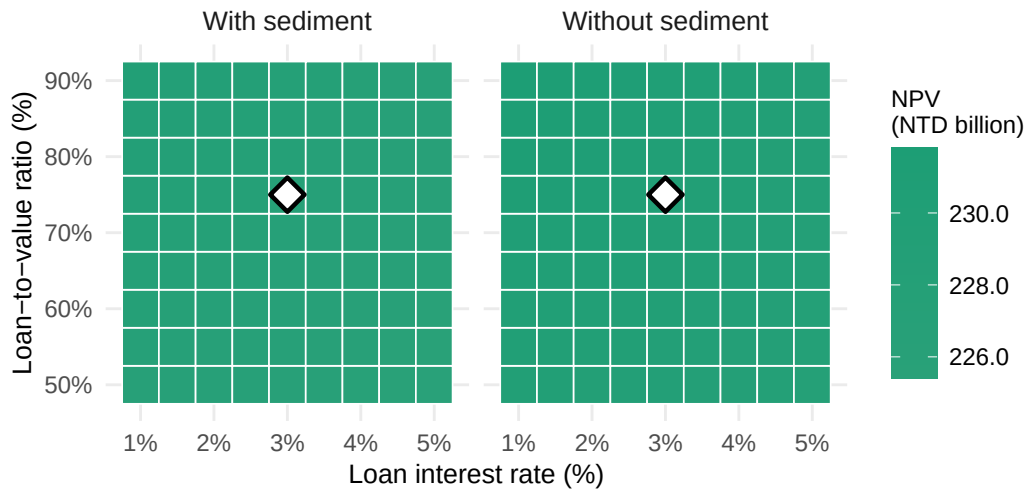
Warning: `stat\_contour()`: Zero contours were generated

Warning in min(x): no non-missing arguments to min; returning Inf

Warning in max(x): no non-missing arguments to max; returning -Inf

NPV Feasibility Frontier

Dashed white line = NPV breakeven (= 0); green = viable, red = loss



Diamond = current parameter setting (LTV = 75%, rate = 3%)
CAPEX = NTD 9.7B; FIT = NTD 2.8599/kWh; Equity rate = 8%

Immediate-dispatch reference

```
sim_forward24 <- run_reservoir_simulation_forward24(  
  daily_flow = daily_flow,  
  e_flow_mode = "recommended",  
  scale_w2 = TRUE  
)  
  
annual_forward24 <- summarise_annual_generation(sim_forward24)  
  
annual_forward24 |>  
  summarise(  
    mean_energy_gwh = mean(energy_gwh_total, na.rm = TRUE),  
    p10_energy_gwh = quantile(energy_gwh_total, 0.10, na.rm = TRUE),  
    p90_energy_gwh = quantile(energy_gwh_total, 0.90, na.rm = TRUE),  
    mean_revenue_b_ntd = mean(revenue_ntd, na.rm = TRUE) / 1e9,
```

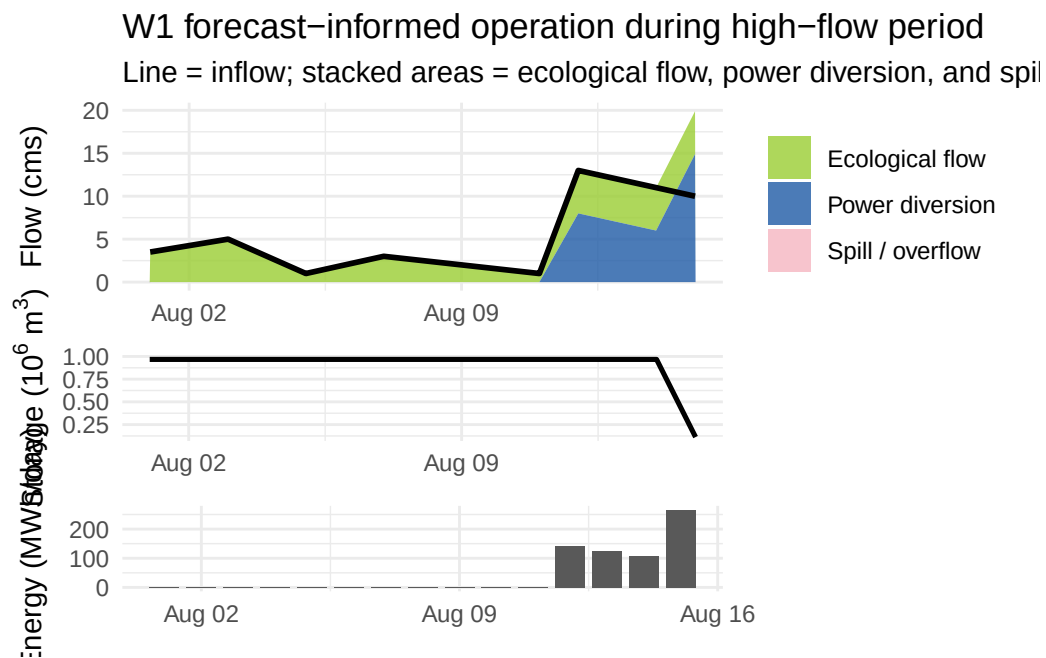
```
mean_spill_m3 = mean(spill_m3_W1 + spill_m3_W2, na.rm = TRUE)
)
```

A tibble: 1 x 5

	mean_energy_gwh	p10_energy_gwh	p90_energy_gwh	mean_revenue_b_ntd	mean_spill_m3
	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	48.2	31.0	62.0	0.138	95554897.

Forecast-Informed Operation Window

```
plot_operation_window(
  sim_df = sim_forward24$W1,
  start_date = "2010-08-01",
  end_date = "2010-08-15",
  title_text = "W1 forecast-informed operation during high-flow period"
)
```



Immediate Dispatch Simulation

```
sim_instant <- run_reservoir_simulation(  
  daily_flow = daily_flow,  
  e_flow_mode = "recommended"  
)
```

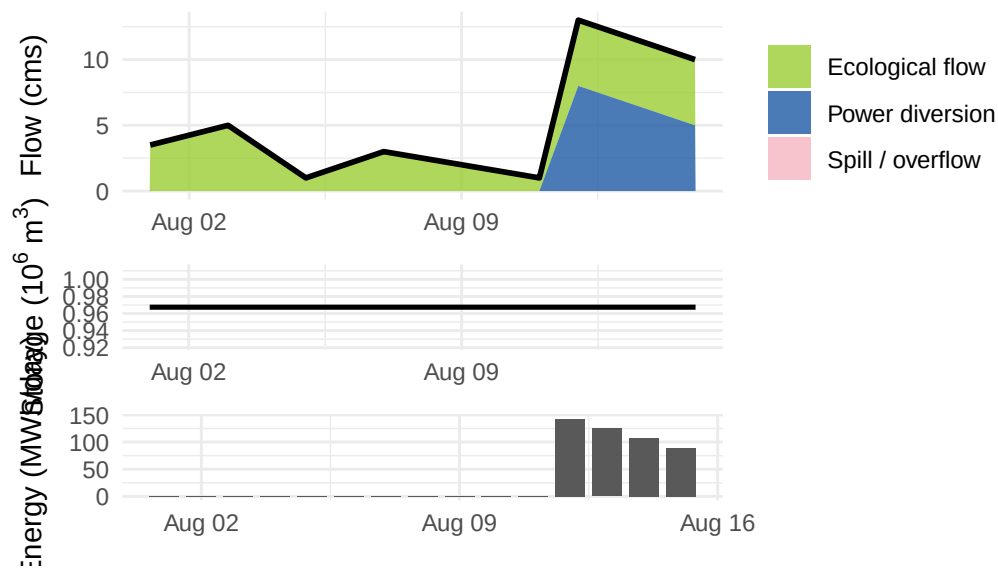
simulate_one_weir [W1]: net_head estimated as 86.3 m

simulate_one_weir [W2]: net_head estimated as 349.4 m

```
plot_operation_window(  
  sim_df      = sim_instant$W1,  
  start_date  = "2010-08-01",  
  end_date    = "2010-08-15",  
  title_text  = "W1 immediate-dispatch operation during high-flow period"  
)
```

W1 immediate-dispatch operation during high-flow period

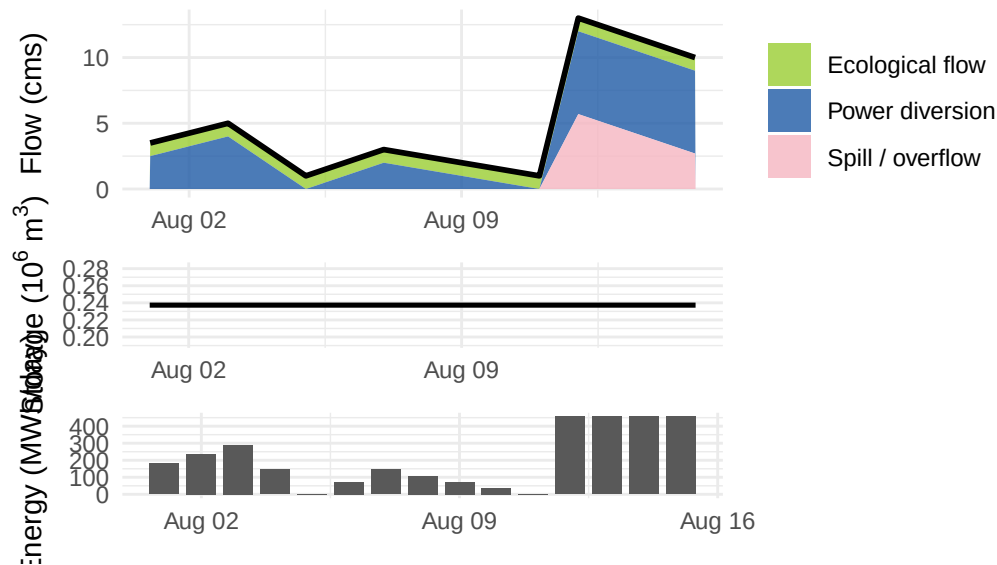
Line = inflow; stacked areas = ecological flow, power diversion, and spill



```
plot_operation_window(
  sim_df      = sim_instant$W2,
  start_date  = "2010-08-01",
  end_date    = "2010-08-15",
  title_text  = "W2 immediate-dispatch operation during high-flow period"
)
```

W2 immediate-dispatch operation during high-flow period

Line = inflow; stacked areas = ecological flow, power diversion, and spill



Annual generation summary — immediate dispatch

```
annual_instant <- sim_instant$combined |>
  mutate(year = lubridate::year(date)) |>
  group_by(year) |>
  summarise(
    energy_gwh_W1 = sum(energy_kwh_W1, na.rm = TRUE) / 1e6,
    energy_gwh_W2 = sum(energy_kwh_W2, na.rm = TRUE) / 1e6,
    energy_gwh_total = sum(energy_kwh_total, na.rm = TRUE) / 1e6,
    revenue_ntd = sum(energy_kwh_total, na.rm = TRUE) * .FIN$fit_ntd_per_kwh,
    .groups = "drop"
  )
```

20-year NPV trajectories

```
npv_forward24_20yr <- run_npv_20yr_from_generation(  
  annual_generation_df = annual_forward24,  
  ltv                   = LTV,  
  r_loan                = R_LOAN,  
  discount_rate         = R_EQUITY  
)  
  
npv_forward24_20yr <- run_npv_20yr_from_generation(  
  annual_generation_df = annual_forward24,  
  ltv = 0.75,  
  r_loan = 0.03,  
  discount_rate = 0.08  
)  
  
npv_instant_20yr <- run_npv_20yr_from_generation(  
  annual_generation_df = annual_instant,  
  ltv                   = LTV,  
  r_loan                = R_LOAN,  
  discount_rate         = R_EQUITY  
)  
  
plot_npv_financial_line_20yr(  
  npv_forward24_20yr,  
  title_text = "Forward-looking operation: 20-year NPV trajectory"  
)
```

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

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conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
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conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

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conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
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Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

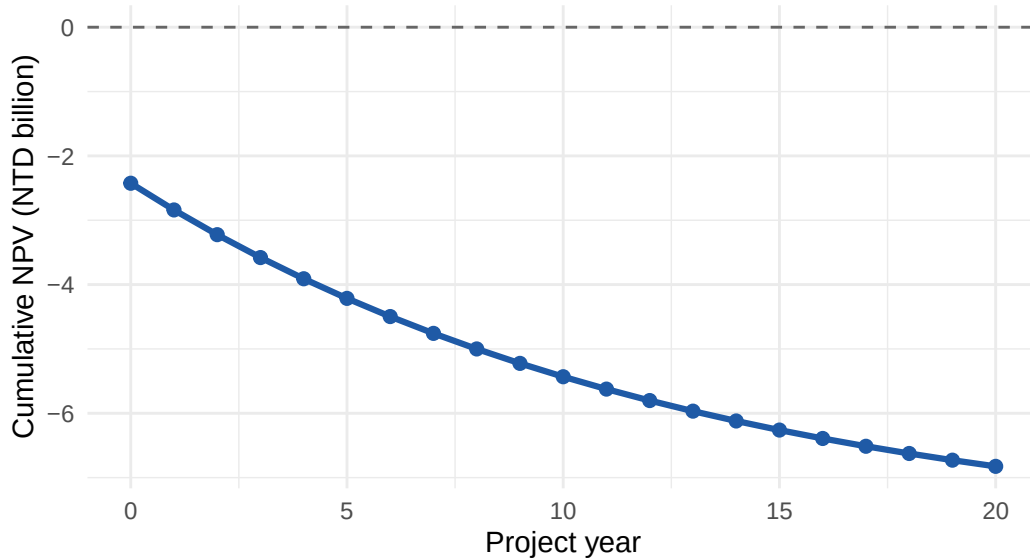
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>


```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>
```

Forward-looking operation: 20-year NPV trajectory

Year 0 includes equity CAPEX; years 1...20 show cumulative discounted ca



```
plot_npv_financial_line_20yr(
  npv_instant_20yr,
  title_text = "Immediate-dispatch operation: 20-year NPV trajectory"
)
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>
```

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

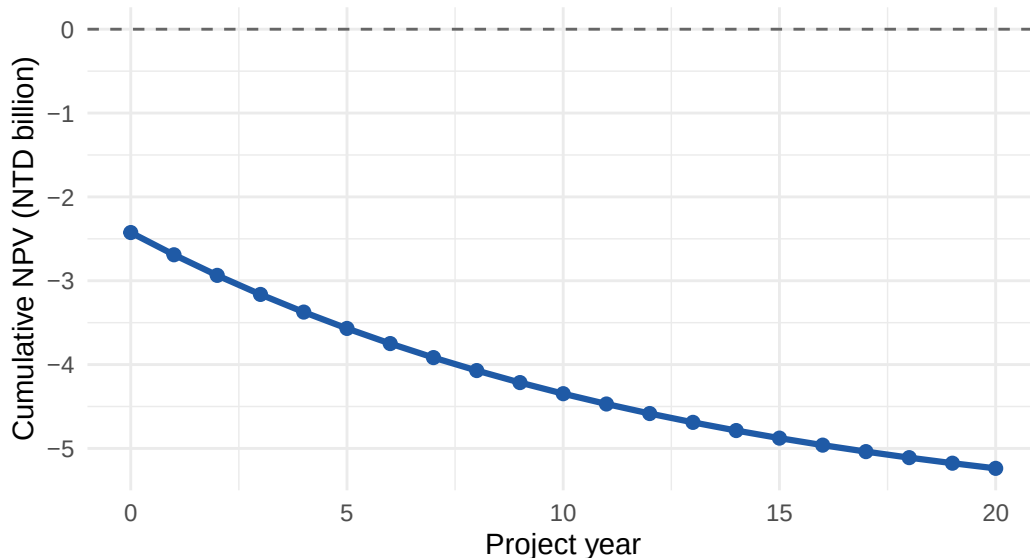
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

```
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>
```

Immediate-dispatch operation: 20-year NPV trajectory

Year 0 includes equity CAPEX; years 1...20 show cumulative discounted cash



```
plot_npv_comparison_20yr(
  npv_instant      = npv_instant_20yr,
  npv_forward24    = npv_forward24_20yr,
  title_text       = "Immediate vs forecast-informed operation: 20-year NPV"
)
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>
```

```
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x$label), x$x, x$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>
```

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

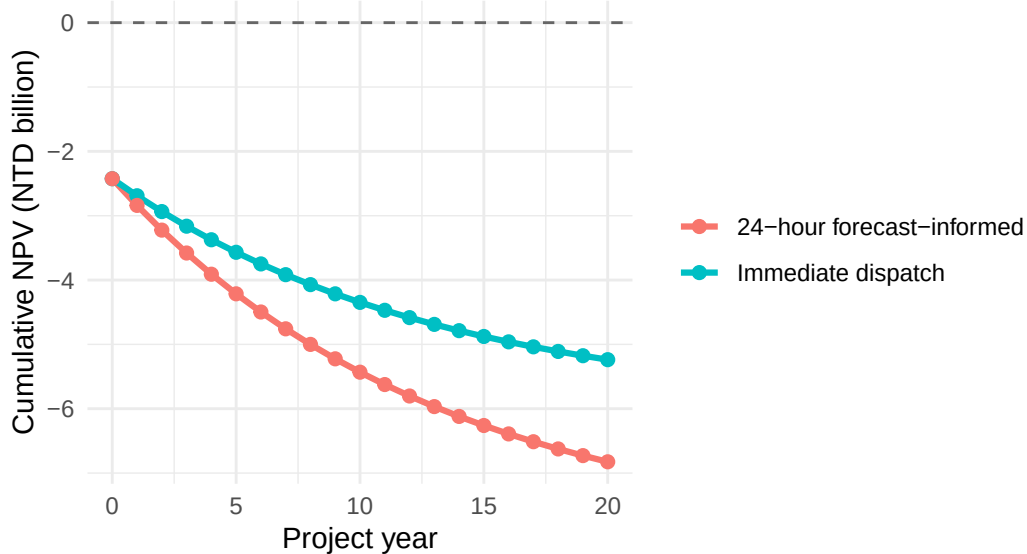
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Immediate vs forecast-informed operation: 20-year NPV

Year 0 includes equity CAPEX; years 1-20 show cumulative discounted cash



Feasibility frontier — operations

```
frontier_instant <- run_sensitivity(
  annual_revenue_ntd = mean(annual_instant$revenue_ntd, na.rm = TRUE),
  r_loan_seq        = seq(0.01, 0.08, by = 0.005),
  ltv_seq           = seq(0.40, 0.90, by = 0.05),
  scenario_label    = "Immediate dispatch"
)

frontier_forward24 <- run_sensitivity(
  annual_revenue_ntd = mean(annual_forward24$revenue_ntd, na.rm = TRUE),
  r_loan_seq        = seq(0.01, 0.08, by = 0.005),
  ltv_seq           = seq(0.40, 0.90, by = 0.05),
  scenario_label    = "24-hour forecast-informed"
)

frontier_compare <- bind_rows(frontier_instant, frontier_forward24)
```

```
plot_sensitivity_frontier(
  frontier_compare,
  title_text = "Immediate vs forecast-informed operation: feasibility frontier"
)
```

Warning: `stat\_contour()`: Zero contours were generated

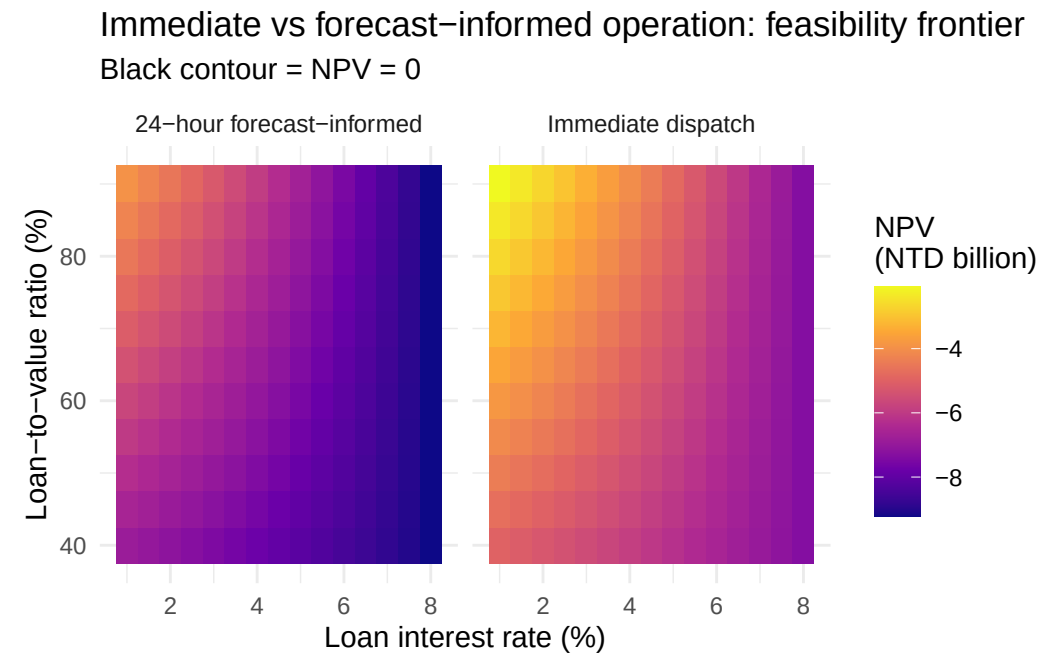
Warning in min(x): no non-missing arguments to min; returning Inf

Warning in max(x): no non-missing arguments to max; returning -Inf

Warning: `stat\_contour()`: Zero contours were generated

Warning in min(x): no non-missing arguments to min; returning Inf

Warning in max(x): no non-missing arguments to max; returning -Inf



20-year NPV trajectories


```

plot_npv_comparison_20yr(
  npv_instant = npv_instant_20yr,
  npv_forward24 = npv_forward24_20yr,
  title_text = "Immediate vs forecast-informed operation: 20-year NPV"
)

```

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
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conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

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conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

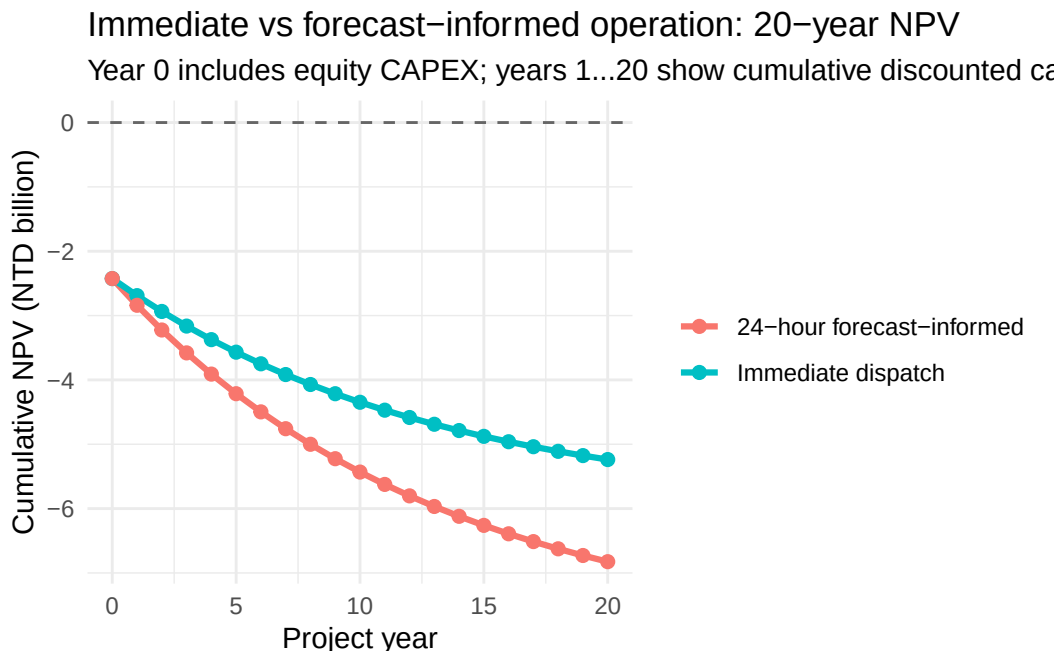
Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

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Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
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conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>



```

npv_forward24_20yr <- run_npv_20yr_from_generation(
  annual_generation_df = annual_forward24,
  ltv = 0.75,
  r_loan = 0.03,
  discount_rate = 0.08
)

plot_npv_financial_line_20yr(
  npv_forward24_20yr,
  title_text = "Forward-looking operation: 20-year NPV trajectory"
)

```

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
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conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

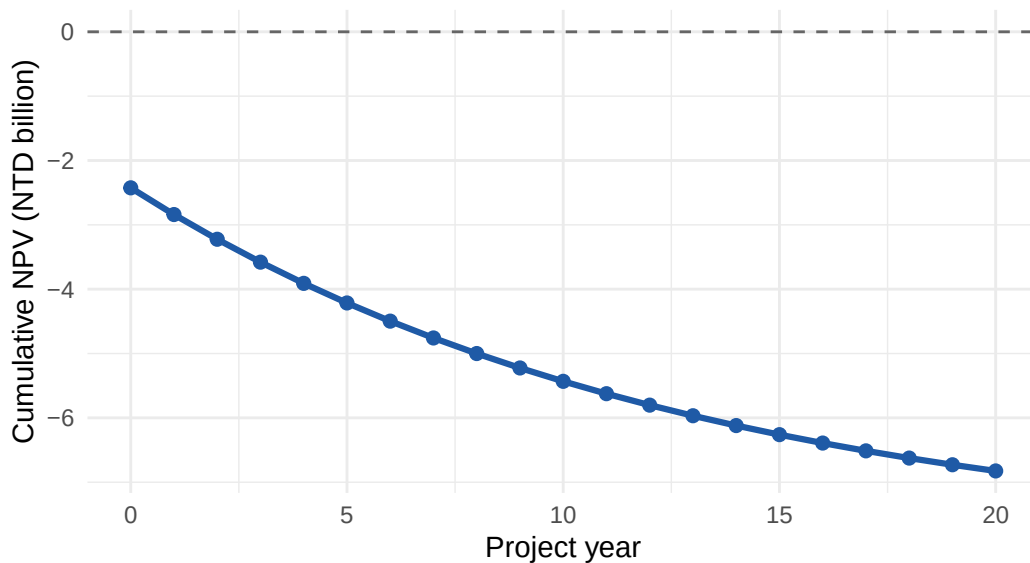
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
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Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Forward-looking operation: 20-year NPV trajectory

Year 0 includes equity CAPEX; years 1...20 show cumulative discounted cash flow



```
npv_instant_20yr <- run_npv_20yr_from_generation(
  annual_generation_df = annual_instant,
  ltv = 0.75,
  r_loan = 0.03,
  discount_rate = 0.08
)

plot_npv_financial_line_20yr(
  npv_instant_20yr,
  title_text = "Immediate-dispatch operation: 20-year NPV trajectory"
)
```

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
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conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
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conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
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conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
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Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
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conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
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conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <e2>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <80>

Warning in grid.Call(C_textBounds, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

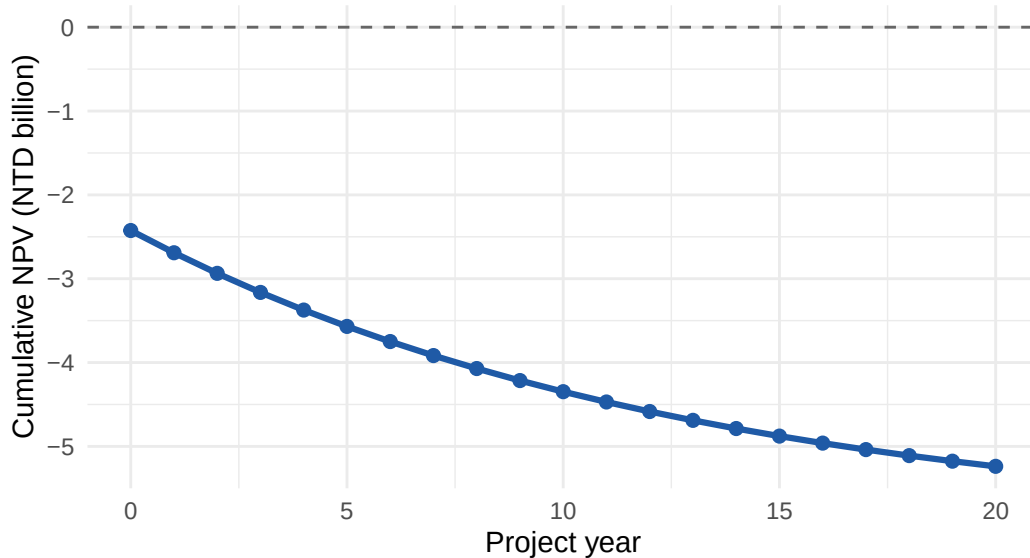
Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
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Warning in grid.Call.graphics(C_text, as.graphicsAnnot(x\$label), x\$x, x\$y, :
conversion failure on 'Year 0 includes equity CAPEX; years 1-20 show cumulative
discounted cash flow' in 'mbcsToSbcs': dot substituted for <93>

Immediate-dispatch operation: 20-year NPV trajectory

Year 0 includes equity CAPEX; years 1-20 show cumulative discounted cash



Operations Feasibility Frontier - Comparing

```
frontier_instant <- run_sensitivity(
  annual_revenue_ntd = mean(annual_instant$revenue_ntd, na.rm = TRUE),
  r_loan_seq = seq(0.01, 0.08, by = 0.005),
  ltv_seq = seq(0.40, 0.90, by = 0.05),
  scenario_label = "Immediate dispatch"
)

frontier_forward24 <- run_sensitivity(
  annual_revenue_ntd = mean(annual_forward24$revenue_ntd, na.rm = TRUE),
  r_loan_seq = seq(0.01, 0.08, by = 0.005),
  ltv_seq = seq(0.40, 0.90, by = 0.05),
  scenario_label = "24-hour forecast-informed"
)

frontier_compare <- bind_rows(
```

```

    frontier_instant,
    frontier_forward24
)

plot_sensitivity_frontier(
    frontier_compare,
    title_text = "Immediate vs forecast-informed operation: feasibility frontier"
)

```

Warning: `stat\_contour()`: Zero contours were generated

Warning in min(x): no non-missing arguments to min; returning Inf

Warning in max(x): no non-missing arguments to max; returning -Inf

Warning: `stat\_contour()`: Zero contours were generated

Warning in min(x): no non-missing arguments to min; returning Inf

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